General Mathematics — Exercise Sheet 3

Sequence limits and their algebra

Variant: Statements

Exercise 1 (Cancellation inside a square root). For $n \in \mathbb{N}$, let $a_n = \sqrt{n^2 + 4n} - n$. Prove directly from the epsilon–N definition that $a_n \to 2$. Give an explicit cutoff for each $\varepsilon > 0$. Use Lecture 2's Archimedean property and Lecture 3's definition.

Exercise 2 (Recovering a factor). Suppose (a_n) and (b_n) are real sequences, $a_n b_n \to L$ and $b_n \to b \neq 0$. Prove $a_n \to L/b$ using Lecture 3's limit laws. Give a counterexample if the condition $b \neq 0$ is removed, with a_n bounded but divergent and $a_n b_n \to 0$. Write the negation of convergence to a proposed $L \in \mathbb{R}$ and use it to justify the nonconvergence of your bounded factor.

Exercise 3 (Interleaving two convergent sequences). Let (a_n) be a real sequence and $L \in \mathbb{R}$. Prove that $a_n \to L$ if and only if both $a_{2k} \to L$ and $a_{2k-1} \to L$ as $k \to \infty$. Use the epsilon definition and Lecture 1's even/odd alternatives, rather than a later subsequence theorem. Apply your criterion when $a_{2k} = 2 + 1/k$ and $a_{2k-1} = 2 - 1/k^2$.

Exercise 4 (Limits of finite geometric sums). For $q \in \mathbb{R}$, let $s_n = q + q^2 + \cdots + q^n$. Prove $(1 - q)s_n = q - q^{n+1}$ by cancellation, and compute $\lim s_n$ when $|q| < 1$. Then explain what happens at $q = 1$ and for $q > 1$. Use only Lecture 3's geometric limit and limit laws, together with Lecture 2's Bernoulli inequality where needed.